In [1]:
```
# Magnificent 7 return baskets: Mag7, Mag7B and Mag7C (Section 3.2.3)
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

# The clean files are saved straight into the folder with the regression notebooks
OUT = "../Regressions and relevant csvs/"
```

In [2]:
```
# Compustat Security Monthly (WRDS): monthly total return (trt1m, in percent) of the seven
stocks,
# December 2015 to December 2025
raw = pd.read_csv("Mag7 Monthly Returns.csv")
stocks = pd.DataFrame({
    "ticker": raw["(tic) Ticker Symbol"],
    "date": pd.to_datetime(raw["(datadate) Data Date - Security Monthly"],
format="%d.%m.%Y"),
    "ret": raw["(trt1m) Monthly Total Return"] / 100,
})
stocks["log_ret"] = np.log(1 + stocks["ret"])

# One column per stock
wide = stocks.pivot(index="date", columns="ticker", values="log_ret")
print(len(wide), "months, missing values:", wide.isna().sum().sum())
```

```
121 months, missing values: 0
```

In [3]:
```
# Equal-weighted baskets of the log returns
mag7 = ["AAPL", "AMZN", "GOOGL", "META", "MSFT", "NVDA", "TSLA"]
mag7b = ["AAPL", "AMZN", "GOOGL", "META", "MSFT", "NVDA"]      # without Tesla
mag7c = ["GOOGL", "META", "MSFT", "NVDA"]                      # AI-focused core

baskets = pd.DataFrame({
    "mag7_ret": wide[mag7].mean(axis=1),
    "mag7b_ret": wide[mag7b].mean(axis=1),
    "mag7c_ret": wide[mag7c].mean(axis=1),
}).reset_index()
baskets.describe()
```

Out[3]:

	date	mag7_ret	mag7b_ret	mag7c_ret
count	121	121.000000	121.000000	121.000000
mean	2020-12-29 23:24:17.851239	0.022908	0.022074	0.024312
min	2015-12-31 00:00:00	-0.197029	-0.194346	-0.198178
25%	2018-06-30 00:00:00	-0.021196	-0.013420	-0.012322
50%	2020-12-31 00:00:00	0.026927	0.029586	0.028648
75%	2023-06-30 00:00:00	0.075163	0.069296	0.073412
max	2025-12-31 00:00:00	0.200230	0.161002	0.164388
std	NaN	0.073652	0.068117	0.072338

In [4]:
```
m = baskets.set_index("date")["mag7_ret"]
print("Largest gains:", ", ".join(f"{v:+.1%} ({d:%b %Y})" for d, v in m.nlargest(3).items()))
print("Largest losses:", ", ".join(f"{v:+.1%} ({d:%b %Y})" for d, v in
m.nsmallest(2).items()))
baskets.drop(columns="date").corr().round(3)
```

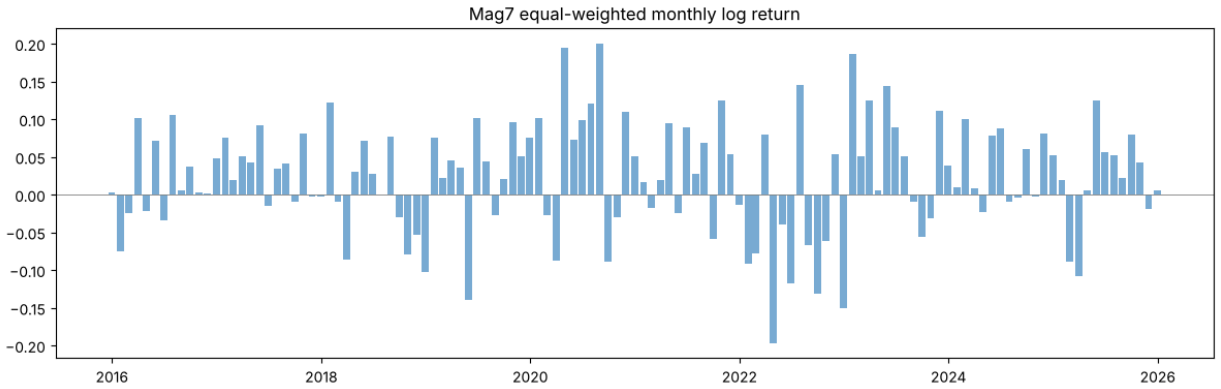
```
Largest gains: +20.0% (Aug 2020), +19.5% (Apr 2020), +18.6% (Jan 2023)
Largest losses: -19.7% (Apr 2022), -15.0% (Dec 2022)
```

Out[4]:

	mag7_ret	mag7b_ret	mag7c_ret
mag7_ret	1.000	0.959	0.914
mag7b_ret	0.959	1.000	0.974
mag7c_ret	0.914	0.974	1.000

In [5]:
```
plt.figure(figsize=(12, 4))
plt.bar(baskets["date"], baskets["mag7_ret"], width=25, alpha=0.6)
plt.axhline(0, color="grey", lw=0.6)
```

```
plt.title("Mag7 equal-weighted monthly log return")
plt.tight_layout()
plt.show()
```



In [6]:
```
baskets[["date", "mag7_ret"]].to_csv(OUT + "mag7_clean.csv", index=False)
baskets[["date", "mag7b_ret"]].to_csv(OUT + "mag7b_clean.csv", index=False)
baskets[["date", "mag7c_ret"]].to_csv(OUT + "mag7c_clean.csv", index=False)
```